Linear Algebra

- Introduction to systems of linear equations:

$ax+by \cancel{+c} = c$ (a, b not both 0)

$ax+by+cz=d$ (a, b, c not all zero)

$a_1x_1+a_2x_2+ \cdots + a_nx_n = b$ — (I)

$a_1x_1+a_2x_2+ \cdots + a_nx_n = 0$ — (II) → a homogeneous linear equation bez R.H.S is zero

Def:

A finite set of linear equation is called a system of linear equations or linear system.

ex:

$$\left.\begin{aligned} 5x+y&=3 \\ 2x-y&=4 \end{aligned}\right\} (5) \qquad \left.\begin{aligned} 4x_1-x_2+3x_3&=-1 \\ 3x_1+x_2+4x_3&=-4 \end{aligned}\right\} (6)$$

$x-y=1$, $2x+y=6$ | $\left.\begin{aligned} x&=\frac{7}{3} \\ y&=\frac{4}{3} \end{aligned}\right\}$ → a linear system with one solution

$x+y=4$, $3x+3y=6$ } There is no solution

$4x-2y=1$, $16x-8y=4$ } That's why there is infinitely many solutions

here $0=0$

$10x-2y=5$ → This equation has also many infinitely solutions

$4x - 2y = 1$

$x = \frac{1}{4} + \frac{1}{2}y$

if $y = t$ then $x = \frac{1}{4} + \frac{1}{2}t$

This value is arbitrary
(Parameter)